

Study Report



Analyzed File	Y_Z_axis v36
Version	Autodesk Fusion (2702.1.58)
Creation Date	2026-05-06, 21:43:14
Author	mohan

☐ **Report Properties**

Title	Studies
Author	mohan

Simulation Model 1

Study 9 - Static Stress

Study Properties

Study Type	Static Stress
Last Modification Date	2026-05-06, 21:25:54

Settings

General

Contact Tolerance	0.10 mm
Remove Rigid Body Modes	No

Mesh

Average Element Size (% of model size)	
Solids	10
Scale Mesh Size Per Part	No
Average Element Size (absolute value)	-
Element Order	Parabolic
Create Curved Mesh Elements	No
Max. Turn Angle on Curves (Deg.)	60
Max. Adjacent Mesh Size Ratio	1.5
Max. Aspect Ratio	10
Minimum Element Size (% of average size)	20

Adaptive Mesh Refinement

Number of Refinement Steps	0
Results Convergence Tolerance (%)	20
Portion of Elements to Refine (%)	10
Results for Baseline Accuracy	von Mises Stress

Materials

Component	Material	Safety Factor
X-axis v8:1/assem2 v5:2/Frame 2 v1:2	Steel	Yield Strength
X-axis v8:1/assem2 v5:2/rail1 Copy v4:2	Steel	Yield Strength
X-axis v8:1/assem2 v5:2/ball bearing v2:2/Body1	Steel	Yield Strength
X-axis v8:1/assem2 v5:2/ball bearing v2:2/Body2	Steel	Yield Strength
X-axis v8:1/assem2 v5:2/ball bearing v2:2/Body3	Steel	Yield Strength
X-axis v8:1/assem2 v5:2/ball bearing v2:2/Body4	Steel	Yield Strength
X-axis v8:1/assem2 v5:2/ball bearing v2:2/Body5	Steel	Yield Strength
X-axis v8:1/assem2 v5:2/ball bearing v2:2/Body6	Steel	Yield Strength
X-axis v8:1/assem2 v5:2/ball bearing v2:2/Body7	Steel	Yield Strength
X-axis v8:1/assem2 v5:2/ball bearing v2:2/Body8	Steel	Yield Strength
X-axis v8:1/assem2 v5:2/ball bearing v2:2/Body9	Steel	Yield Strength
X-axis v8:1/assem2 v5:2/ball bearing v2:2/Body10	Steel	Yield Strength
X-axis v8:1/assem2 v5:2/ball bearing v2:2/Body11	Steel	Yield Strength
X-axis v8:1/assem2 v5:2/ball bearing v2:2/Body12	Steel	Yield Strength
X-axis v8:1/assem2 v5:2/bolt v1:12	Steel	Yield Strength
X-axis v8:1/assem2 v5:2/bolt v1:13	Steel	Yield Strength
X-axis v8:1/assem2 v5:2/bolt v1:14	Steel	Yield Strength

X-axis v8:1/assem2 v5:2/bolt v1:15	Steel	Yield Strength
X-axis v8:1/assem2 v5:2/bolt v1:16	Steel	Yield Strength
X-axis v8:1/assem2 v5:2/bolt v1:2	Steel	Yield Strength
X-axis v8:1/assem2 v5:2/bolt v1:10	Steel	Yield Strength
X-axis v8:1/assem2 v5:2/bolt v1:11	Steel	Yield Strength
X-axis v8:1/assem2 v5:2/Frame 1 v1:2	Steel	Yield Strength
X-axis v8:1/assem2 v5:2/rod v1:4	Stainless Steel, 440C	Yield Strength
X-axis v8:1/assem2 v5:2/rod v1:2	Stainless Steel, 440C	Yield Strength
X-axis v8:1/slider v4:2	Stainless Steel, 440C	Yield Strength
X-axis v8:1/Frame 1.2 v1:3	Steel	Yield Strength
X-axis v8:1/ball bearing v2:4/Body1	Steel	Yield Strength
X-axis v8:1/ball bearing v2:4/Body2	Steel	Yield Strength
X-axis v8:1/ball bearing v2:4/Body3	Steel	Yield Strength
X-axis v8:1/ball bearing v2:4/Body4	Steel	Yield Strength
X-axis v8:1/ball bearing v2:4/Body5	Steel	Yield Strength
X-axis v8:1/ball bearing v2:4/Body6	Steel	Yield Strength
X-axis v8:1/ball bearing v2:4/Body7	Steel	Yield Strength
X-axis v8:1/ball bearing v2:4/Body8	Steel	Yield Strength
X-axis v8:1/ball bearing v2:4/Body9	Steel	Yield Strength
X-axis v8:1/ball bearing v2:4/Body10	Steel	Yield Strength
X-axis v8:1/ball bearing v2:4/Body11	Steel	Yield Strength
X-axis v8:1/ball bearing v2:4/Body12	Steel	Yield Strength
X-axis v8:1/coupler v1:3	Steel	Yield Strength
X-axis v8:1/traveller v8:2	Stainless Steel, 440C	Yield Strength
X-axis v8:1/motor v2:2/base:1	Stainless Steel, 440C	Yield Strength
X-axis v8:1/motor v2:2/middle:1	Stainless Steel, 440C	Yield Strength
X-axis v8:1/motor v2:2/top:1	Stainless Steel, 440C	Yield Strength
X-axis v8:1/motor v2:2/rod:1	Stainless Steel, 440C	Yield Strength
Flange for worktable v3:1	Steel	Yield Strength
Flange for worktable v3:2	Steel	Yield Strength
Component1:1	Steel	Yield Strength
Conical Spring Washer JIS B 1252 - 6 - 2H Steel 100 HV Plain v1:5	Steel 100 HV, Plain	Yield Strength
Conical Spring Washer JIS B 1252 - 6 - 2H Steel 100 HV Plain v1:6	Steel 100 HV, Plain	Yield Strength
assem4 v18:1/assem 3 v10:1/Frame 1.2 v1:4	Steel	Yield Strength
assem4 v18:1/assem 3 v10:1/assem2 v5:3/Frame 2 v1:2	Steel	Yield Strength
assem4 v18:1/assem 3 v10:1/assem2 v5:3/rail1 Copy v4:2	Steel	Yield Strength
assem4 v18:1/assem 3 v10:1/assem2 v5:3/ball bearing v2:2/Body1	Steel	Yield Strength
assem4 v18:1/assem 3 v10:1/assem2 v5:3/ball bearing v2:2/Body2	Steel	Yield Strength
assem4 v18:1/assem 3 v10:1/assem2 v5:3/ball bearing v2:2/Body3	Steel	Yield Strength
assem4 v18:1/assem 3 v10:1/assem2 v5:3/ball bearing v2:2/Body4	Steel	Yield Strength
assem4 v18:1/assem 3 v10:1/assem2 v5:3/ball bearing v2:2/Body5	Steel	Yield Strength
assem4 v18:1/assem 3 v10:1/assem2 v5:3/ball bearing v2:2/Body6	Steel	Yield Strength
assem4 v18:1/assem 3 v10:1/assem2 v5:3/ball bearing v2:2/Body7	Steel	Yield Strength
assem4 v18:1/assem 3 v10:1/assem2 v5:3/ball bearing v2:2/Body8	Steel	Yield Strength
assem4 v18:1/assem 3 v10:1/assem2 v5:3/ball bearing v2:2/Body9	Steel	Yield Strength
assem4 v18:1/assem 3 v10:1/assem2 v5:3/ball bearing v2:2/Body10	Steel	Yield Strength
assem4 v18:1/assem 3 v10:1/assem2 v5:3/ball bearing v2:2/Body11	Steel	Yield Strength
assem4 v18:1/assem 3 v10:1/assem2 v5:3/ball bearing v2:2/Body12	Steel	Yield Strength
assem4 v18:1/assem 3 v10:1/assem2 v5:3/bolt v1:12	Steel	Yield Strength
assem4 v18:1/assem 3 v10:1/assem2 v5:3/bolt v1:13	Steel	Yield Strength
assem4 v18:1/assem 3 v10:1/assem2 v5:3/bolt v1:14	Steel	Yield Strength
assem4 v18:1/assem 3 v10:1/assem2 v5:3/bolt v1:15	Steel	Yield Strength
assem4 v18:1/assem 3 v10:1/assem2 v5:3/bolt v1:16	Steel	Yield Strength

assem4 v18:1/assem 3 v10:1/assem2 v5:3/bolt v1:2	Steel	Yield Strength
assem4 v18:1/assem 3 v10:1/assem2 v5:3/bolt v1:10	Steel	Yield Strength
assem4 v18:1/assem 3 v10:1/assem2 v5:3/bolt v1:11	Steel	Yield Strength
assem4 v18:1/assem 3 v10:1/assem2 v5:3/Frame 1 v1:2	Steel	Yield Strength
assem4 v18:1/assem 3 v10:1/assem2 v5:3/rod v1:4	Stainless Steel, 440C	Yield Strength
assem4 v18:1/assem 3 v10:1/assem2 v5:3/rod v1:2	Stainless Steel, 440C	Yield Strength
assem4 v18:1/assem 3 v10:1/ball bearing v2:5/Body1	Steel	Yield Strength
assem4 v18:1/assem 3 v10:1/ball bearing v2:5/Body2	Steel	Yield Strength
assem4 v18:1/assem 3 v10:1/ball bearing v2:5/Body3	Steel	Yield Strength
assem4 v18:1/assem 3 v10:1/ball bearing v2:5/Body4	Steel	Yield Strength
assem4 v18:1/assem 3 v10:1/ball bearing v2:5/Body5	Steel	Yield Strength
assem4 v18:1/assem 3 v10:1/ball bearing v2:5/Body6	Steel	Yield Strength
assem4 v18:1/assem 3 v10:1/ball bearing v2:5/Body7	Steel	Yield Strength
assem4 v18:1/assem 3 v10:1/ball bearing v2:5/Body8	Steel	Yield Strength
assem4 v18:1/assem 3 v10:1/ball bearing v2:5/Body9	Steel	Yield Strength
assem4 v18:1/assem 3 v10:1/ball bearing v2:5/Body10	Steel	Yield Strength
assem4 v18:1/assem 3 v10:1/ball bearing v2:5/Body11	Steel	Yield Strength
assem4 v18:1/assem 3 v10:1/ball bearing v2:5/Body12	Steel	Yield Strength
assem4 v18:1/assem 3 v10:1/slider v4:3	Stainless Steel, 440C	Yield Strength
assem4 v18:1/assem 3 v10:1/motor v2:3/base:1	Steel	Yield Strength
assem4 v18:1/assem 3 v10:1/motor v2:3/middle:1	Steel	Yield Strength
assem4 v18:1/assem 3 v10:1/motor v2:3/top:1	Steel	Yield Strength
assem4 v18:1/assem 3 v10:1/motor v2:3/rod:1	Steel	Yield Strength
assem4 v18:1/assem 3 v10:1/coupler v1:4	Steel	Yield Strength
assem4 v18:1/assem 3 v10:1/traveller v8:3	Steel	Yield Strength

Steel

Density	7.850E-06 kg / mm ³
Young's Modulus	210000.00 MPa
Poisson's Ratio	0.30
Yield Strength	207.00 MPa
Ultimate Tensile Strength	345.00 MPa
Thermal Conductivity	0.056 W / (mm C)
Thermal Expansion Coefficient	1.200E-05 / C
Specific Heat	480.00 J / (kg C)

Steel 100 HV, Plain

Density	8.030E-06 kg / mm ³
Young's Modulus	195000.00 MPa
Poisson's Ratio	0.29
Yield Strength	450.00 MPa
Ultimate Tensile Strength	700.00 MPa
Thermal Conductivity	0.016 W / (mm C)
Thermal Expansion Coefficient	1.730E-05 / C
Specific Heat	500.00 J / (kg C)

Stainless Steel, 440C

Density	7.750E-06 kg / mm ³
Young's Modulus	206700.00 MPa
Poisson's Ratio	0.27
Yield Strength	689.00 MPa
Ultimate Tensile Strength	861.25 MPa

Thermal Conductivity	0.024 W / (mm C)
Thermal Expansion Coefficient	1.040E-05 / C
Specific Heat	460.00 J / (kg C)

☐ Contacts

☐ Bonded

Name
[S] Bonded39 [top:1 rod:1]
[S] Bonded40 [middle:1 top:1]
[S] Bonded41 [middle:1 top:1]
[S] Bonded42 [middle:1 top:1]
[S] Bonded43 [middle:1 top:1]
[S] Bonded44 [middle:1 rod:1]
[S] Bonded45 [middle:1 top:1]
[S] Bonded46 [middle:1 top:1]
[S] Bonded47 [middle:1 top:1]
[S] Bonded48 [middle:1 top:1]
[S] Bonded49 [middle:1 top:1]
[S] Bonded50 [middle:1 top:1]
[S] Bonded51 [middle:1 top:1]
[S] Bonded52 [middle:1 top:1]
[S] Bonded53 [middle:1 top:1]
[S] Bonded54 [base:1 middle:1]
[S] Bonded55 [base:1 middle:1]
[S] Bonded56 [base:1 middle:1]
[S] Bonded57 [base:1 middle:1]
[S] Bonded58 [base:1 middle:1]
[S] Bonded59 [base:1 middle:1]
[S] Bonded60 [base:1 middle:1]
[S] Bonded61 [base:1 middle:1]
[S] Bonded62 [base:1 middle:1]
[S] Bonded63 [base:1 middle:1]
[S] Bonded64 [base:1 middle:1]
[S] Bonded65 [base:1 middle:1]
[S] Bonded66 [base:1 middle:1]
[S] Bonded67 [slider v4:3 traveller v8:3]
[S] Bonded68 [ball bearing v2:5(Body11) ball bearing v2:5(Body12)]
[S] Bonded69 [ball bearing v2:5(Body11) ball bearing v2:5(Body12)]
[S] Bonded70 [ball bearing v2:5(Body11) ball bearing v2:5(Body12)]
[S] Bonded71 [ball bearing v2:5(Body11) ball bearing v2:5(Body12)]
[S] Bonded72 [ball bearing v2:5(Body11) ball bearing v2:5(Body12)]
[S] Bonded73 [ball bearing v2:5(Body11) ball bearing v2:5(Body12)]
[S] Bonded74 [ball bearing v2:5(Body11) ball bearing v2:5(Body12)]
[S] Bonded75 [ball bearing v2:5(Body11) ball bearing v2:5(Body12)]
[S] Bonded76 [ball bearing v2:5(Body10) ball bearing v2:5(Body12)]
[S] Bonded77 [ball bearing v2:5(Body10) ball bearing v2:5(Body11)]
[S] Bonded78 [ball bearing v2:5(Body9) ball bearing v2:5(Body12)]
[S] Bonded79 [ball bearing v2:5(Body9) ball bearing v2:5(Body11)]
[S] Bonded80 [ball bearing v2:5(Body8) ball bearing v2:5(Body12)]
[S] Bonded81 [ball bearing v2:5(Body8) ball bearing v2:5(Body11)]
[S] Bonded82 [ball bearing v2:5(Body7) ball bearing v2:5(Body12)]
[S] Bonded83 [ball bearing v2:5(Body7) ball bearing v2:5(Body11)]

[S] Bonded84 [ball bearing v2:5(Body6) ball bearing v2:5(Body12)]
[S] Bonded85 [ball bearing v2:5(Body6) ball bearing v2:5(Body11)]
[S] Bonded86 [ball bearing v2:5(Body5) ball bearing v2:5(Body12)]
[S] Bonded87 [ball bearing v2:5(Body5) ball bearing v2:5(Body11)]
[S] Bonded88 [ball bearing v2:5(Body4) ball bearing v2:5(Body12)]
[S] Bonded89 [ball bearing v2:5(Body4) ball bearing v2:5(Body11)]
[S] Bonded90 [ball bearing v2:5(Body3) ball bearing v2:5(Body12)]
[S] Bonded91 [ball bearing v2:5(Body3) ball bearing v2:5(Body11)]
[S] Bonded92 [ball bearing v2:5(Body2) ball bearing v2:5(Body10)]
[S] Bonded93 [ball bearing v2:5(Body2) ball bearing v2:5(Body9)]
[S] Bonded94 [ball bearing v2:5(Body2) ball bearing v2:5(Body8)]
[S] Bonded95 [ball bearing v2:5(Body2) ball bearing v2:5(Body7)]
[S] Bonded96 [ball bearing v2:5(Body2) ball bearing v2:5(Body6)]
[S] Bonded97 [ball bearing v2:5(Body2) ball bearing v2:5(Body5)]
[S] Bonded98 [ball bearing v2:5(Body2) ball bearing v2:5(Body4)]
[S] Bonded99 [ball bearing v2:5(Body2) ball bearing v2:5(Body3)]
[S] Bonded100 [ball bearing v2:5(Body1) ball bearing v2:5(Body10)]
[S] Bonded101 [ball bearing v2:5(Body1) ball bearing v2:5(Body9)]
[S] Bonded102 [ball bearing v2:5(Body1) ball bearing v2:5(Body8)]
[S] Bonded103 [ball bearing v2:5(Body1) ball bearing v2:5(Body7)]
[S] Bonded104 [ball bearing v2:5(Body1) ball bearing v2:5(Body6)]
[S] Bonded105 [ball bearing v2:5(Body1) ball bearing v2:5(Body5)]
[S] Bonded106 [ball bearing v2:5(Body1) ball bearing v2:5(Body4)]
[S] Bonded107 [ball bearing v2:5(Body1) ball bearing v2:5(Body3)]
[S] Bonded108 [ball bearing v2:5(Body1) slider v4:3]
[S] Bonded109 [ball bearing v2:5(Body1) slider v4:3]
[S] Bonded110 [rod v1:2 traveller v8:3]
[S] Bonded111 [rod v1:4 traveller v8:3]
[S] Bonded112 [Frame 1 v1:2 rod v1:2]
[S] Bonded113 [Frame 1 v1:2 rod v1:2]
[S] Bonded114 [Frame 1 v1:2 rod v1:4]
[S] Bonded115 [Frame 1 v1:2 rod v1:4]
[S] Bonded116 [bolt v1:11 Frame 1 v1:2]
[S] Bonded117 [bolt v1:11 Frame 1 v1:2]
[S] Bonded118 [bolt v1:11 Frame 1 v1:2]
[S] Bonded119 [bolt v1:10 Frame 1 v1:2]
[S] Bonded120 [bolt v1:10 Frame 1 v1:2]
[S] Bonded121 [bolt v1:10 Frame 1 v1:2]
[S] Bonded122 [bolt v1:2 Frame 1 v1:2]
[S] Bonded123 [bolt v1:2 Frame 1 v1:2]
[S] Bonded124 [bolt v1:2 Frame 1 v1:2]
[S] Bonded125 [bolt v1:16 Frame 1 v1:2]
[S] Bonded126 [bolt v1:16 Frame 1 v1:2]
[S] Bonded127 [bolt v1:16 Frame 1 v1:2]
[S] Bonded128 [ball bearing v2:2(Body11) ball bearing v2:2(Body12)]
[S] Bonded129 [ball bearing v2:2(Body11) ball bearing v2:2(Body12)]
[S] Bonded130 [ball bearing v2:2(Body11) ball bearing v2:2(Body12)]
[S] Bonded131 [ball bearing v2:2(Body11) ball bearing v2:2(Body12)]
[S] Bonded132 [ball bearing v2:2(Body11) ball bearing v2:2(Body12)]
[S] Bonded133 [ball bearing v2:2(Body11) ball bearing v2:2(Body12)]
[S] Bonded134 [ball bearing v2:2(Body11) ball bearing v2:2(Body12)]
[S] Bonded135 [ball bearing v2:2(Body11) ball bearing v2:2(Body12)]
[S] Bonded136 [ball bearing v2:2(Body10) ball bearing v2:2(Body12)]

[S] Bonded137 [ball bearing v2:2(Body10) ball bearing v2:2(Body11)]
[S] Bonded138 [ball bearing v2:2(Body9) ball bearing v2:2(Body12)]
[S] Bonded139 [ball bearing v2:2(Body9) ball bearing v2:2(Body11)]
[S] Bonded140 [ball bearing v2:2(Body8) ball bearing v2:2(Body12)]
[S] Bonded141 [ball bearing v2:2(Body8) ball bearing v2:2(Body11)]
[S] Bonded142 [ball bearing v2:2(Body7) ball bearing v2:2(Body12)]
[S] Bonded143 [ball bearing v2:2(Body7) ball bearing v2:2(Body11)]
[S] Bonded144 [ball bearing v2:2(Body6) ball bearing v2:2(Body12)]
[S] Bonded145 [ball bearing v2:2(Body6) ball bearing v2:2(Body11)]
[S] Bonded146 [ball bearing v2:2(Body5) ball bearing v2:2(Body12)]
[S] Bonded147 [ball bearing v2:2(Body5) ball bearing v2:2(Body11)]
[S] Bonded148 [ball bearing v2:2(Body4) ball bearing v2:2(Body12)]
[S] Bonded149 [ball bearing v2:2(Body4) ball bearing v2:2(Body11)]
[S] Bonded150 [ball bearing v2:2(Body3) ball bearing v2:2(Body12)]
[S] Bonded151 [ball bearing v2:2(Body3) ball bearing v2:2(Body11)]
[S] Bonded152 [ball bearing v2:2(Body2) Frame 1 v1:2]
[S] Bonded153 [ball bearing v2:2(Body2) slider v4:3]
[S] Bonded154 [ball bearing v2:2(Body2) slider v4:3]
[S] Bonded155 [ball bearing v2:2(Body2) ball bearing v2:2(Body10)]
[S] Bonded156 [ball bearing v2:2(Body2) ball bearing v2:2(Body9)]
[S] Bonded157 [ball bearing v2:2(Body2) ball bearing v2:2(Body8)]
[S] Bonded158 [ball bearing v2:2(Body2) ball bearing v2:2(Body7)]
[S] Bonded159 [ball bearing v2:2(Body2) ball bearing v2:2(Body6)]
[S] Bonded160 [ball bearing v2:2(Body2) ball bearing v2:2(Body5)]
[S] Bonded161 [ball bearing v2:2(Body2) ball bearing v2:2(Body4)]
[S] Bonded162 [ball bearing v2:2(Body2) ball bearing v2:2(Body3)]
[S] Bonded163 [ball bearing v2:2(Body1) ball bearing v2:2(Body10)]
[S] Bonded164 [ball bearing v2:2(Body1) ball bearing v2:2(Body9)]
[S] Bonded165 [ball bearing v2:2(Body1) ball bearing v2:2(Body8)]
[S] Bonded166 [ball bearing v2:2(Body1) ball bearing v2:2(Body7)]
[S] Bonded167 [ball bearing v2:2(Body1) ball bearing v2:2(Body6)]
[S] Bonded168 [ball bearing v2:2(Body1) ball bearing v2:2(Body5)]
[S] Bonded169 [ball bearing v2:2(Body1) ball bearing v2:2(Body4)]
[S] Bonded170 [ball bearing v2:2(Body1) ball bearing v2:2(Body3)]
[S] Bonded171 [ball bearing v2:2(Body1) slider v4:3]
[S] Bonded172 [ball bearing v2:2(Body1) slider v4:3]
[S] Bonded173 [rail1 Copy v4:2 Frame 1 v1:2]
[S] Bonded174 [rail1 Copy v4:2 bolt v1:16]
[S] Bonded175 [rail1 Copy v4:2 bolt v1:12]
[S] Bonded176 [rail1 Copy v4:2 bolt v1:2]
[S] Bonded177 [rail1 Copy v4:2 bolt v1:15]
[S] Bonded178 [rail1 Copy v4:2 bolt v1:11]
[S] Bonded179 [rail1 Copy v4:2 bolt v1:13]
[S] Bonded180 [rail1 Copy v4:2 bolt v1:10]
[S] Bonded181 [rail1 Copy v4:2 bolt v1:14]
[S] Bonded182 [Frame 2 v1:2 rail1 Copy v4:2]
[S] Bonded183 [Frame 2 v1:2 top:1]
[S] Bonded184 [Frame 2 v1:2 bolt v1:15]
[S] Bonded185 [Frame 2 v1:2 bolt v1:12]
[S] Bonded186 [Frame 2 v1:2 bolt v1:14]
[S] Bonded187 [Frame 2 v1:2 bolt v1:13]
[S] Bonded188 [Frame 2 v1:2 top:1]
[S] Bonded189 [Frame 2 v1:2 rod v1:4]

[S] Bonded190 [Frame 2 v1:2 rod v1:4]
[S] Bonded191 [Frame 2 v1:2 rod v1:2]
[S] Bonded192 [Frame 2 v1:2 rod v1:2]
[S] Bonded193 [Frame 2 v1:2 bolt v1:15]
[S] Bonded194 [Frame 2 v1:2 bolt v1:15]
[S] Bonded195 [Frame 2 v1:2 bolt v1:12]
[S] Bonded196 [Frame 2 v1:2 bolt v1:12]
[S] Bonded197 [Frame 2 v1:2 bolt v1:14]
[S] Bonded198 [Frame 2 v1:2 bolt v1:14]
[S] Bonded199 [Frame 2 v1:2 bolt v1:13]
[S] Bonded200 [Frame 2 v1:2 bolt v1:13]
[S] Bonded201 [Frame 1.2 v1:4 rail1 Copy v4:2]
[S] Bonded202 [Frame 1.2 v1:4 rail1 Copy v4:2]
[S] Bonded203 [Frame 1.2 v1:4 rail1 Copy v4:2]
[S] Bonded204 [Frame 1.2 v1:4 rail1 Copy v4:2]
[S] Bonded205 [Frame 1.2 v1:4 rail1 Copy v4:2]
[S] Bonded206 [Frame 1.2 v1:4 rod v1:2]
[S] Bonded207 [Frame 1.2 v1:4 rod v1:4]
[S] Bonded208 [Frame 1.2 v1:4 slider v4:3]
[S] Bonded209 [Frame 1.2 v1:4 top:1]
[S] Bonded210 [Frame 1.2 v1:4 Frame 2 v1:2]
[S] Bonded211 [Flange for worktable v3:2 Component1:1]
[S] Bonded212 [Flange for worktable v3:2 Component1:1]
[S] Bonded213 [Flange for worktable v3:1 Component1:1]
[S] Bonded214 [Flange for worktable v3:1 Component1:1]
[S] Bonded215 [Flange for worktable v3:1 Conical Spring Washer JIS B 1252 - 6 - 2H Steel 100 HV Plain v1:6]
[S] Bonded216 [Flange for worktable v3:1 Conical Spring Washer JIS B 1252 - 6 - 2H Steel 100 HV Plain v1:5]
[S] Bonded217 [top:1 rod:1]
[S] Bonded218 [middle:1 top:1]
[S] Bonded219 [middle:1 top:1]
[S] Bonded220 [middle:1 top:1]
[S] Bonded221 [middle:1 top:1]
[S] Bonded222 [middle:1 rod:1]
[S] Bonded223 [middle:1 top:1]
[S] Bonded224 [middle:1 top:1]
[S] Bonded225 [middle:1 top:1]
[S] Bonded226 [middle:1 top:1]
[S] Bonded227 [middle:1 top:1]
[S] Bonded228 [middle:1 top:1]
[S] Bonded229 [middle:1 top:1]
[S] Bonded230 [middle:1 top:1]
[S] Bonded231 [middle:1 top:1]
[S] Bonded232 [base:1 middle:1]
[S] Bonded233 [base:1 middle:1]
[S] Bonded234 [base:1 middle:1]
[S] Bonded235 [base:1 middle:1]
[S] Bonded236 [base:1 middle:1]
[S] Bonded237 [base:1 middle:1]
[S] Bonded238 [base:1 middle:1]
[S] Bonded239 [base:1 middle:1]
[S] Bonded240 [base:1 middle:1]
[S] Bonded241 [base:1 middle:1]
[S] Bonded242 [base:1 middle:1]

[S] Bonded243 [base:1 middle:1]
[S] Bonded244 [base:1 middle:1]
[S] Bonded245 [traveller v8:2 Flange for worktable v3:2]
[S] Bonded246 [traveller v8:2 Component1:1]
[S] Bonded247 [traveller v8:2 Flange for worktable v3:1]
[S] Bonded248 [ball bearing v2:4(Body11) ball bearing v2:4(Body12)]
[S] Bonded249 [ball bearing v2:4(Body11) ball bearing v2:4(Body12)]
[S] Bonded250 [ball bearing v2:4(Body11) ball bearing v2:4(Body12)]
[S] Bonded251 [ball bearing v2:4(Body11) ball bearing v2:4(Body12)]
[S] Bonded252 [ball bearing v2:4(Body11) ball bearing v2:4(Body12)]
[S] Bonded253 [ball bearing v2:4(Body11) ball bearing v2:4(Body12)]
[S] Bonded254 [ball bearing v2:4(Body11) ball bearing v2:4(Body12)]
[S] Bonded255 [ball bearing v2:4(Body11) ball bearing v2:4(Body12)]
[S] Bonded256 [ball bearing v2:4(Body10) ball bearing v2:4(Body12)]
[S] Bonded257 [ball bearing v2:4(Body10) ball bearing v2:4(Body11)]
[S] Bonded258 [ball bearing v2:4(Body9) ball bearing v2:4(Body12)]
[S] Bonded259 [ball bearing v2:4(Body9) ball bearing v2:4(Body11)]
[S] Bonded260 [ball bearing v2:4(Body8) ball bearing v2:4(Body12)]
[S] Bonded261 [ball bearing v2:4(Body8) ball bearing v2:4(Body11)]
[S] Bonded262 [ball bearing v2:4(Body7) ball bearing v2:4(Body12)]
[S] Bonded263 [ball bearing v2:4(Body7) ball bearing v2:4(Body11)]
[S] Bonded264 [ball bearing v2:4(Body6) ball bearing v2:4(Body12)]
[S] Bonded265 [ball bearing v2:4(Body6) ball bearing v2:4(Body11)]
[S] Bonded266 [ball bearing v2:4(Body5) ball bearing v2:4(Body12)]
[S] Bonded267 [ball bearing v2:4(Body5) ball bearing v2:4(Body11)]
[S] Bonded268 [ball bearing v2:4(Body4) ball bearing v2:4(Body12)]
[S] Bonded269 [ball bearing v2:4(Body4) ball bearing v2:4(Body11)]
[S] Bonded270 [ball bearing v2:4(Body3) ball bearing v2:4(Body12)]
[S] Bonded271 [ball bearing v2:4(Body3) ball bearing v2:4(Body11)]
[S] Bonded272 [ball bearing v2:4(Body2) ball bearing v2:4(Body10)]
[S] Bonded273 [ball bearing v2:4(Body2) ball bearing v2:4(Body9)]
[S] Bonded274 [ball bearing v2:4(Body2) ball bearing v2:4(Body8)]
[S] Bonded275 [ball bearing v2:4(Body2) ball bearing v2:4(Body7)]
[S] Bonded276 [ball bearing v2:4(Body2) ball bearing v2:4(Body6)]
[S] Bonded277 [ball bearing v2:4(Body2) ball bearing v2:4(Body5)]
[S] Bonded278 [ball bearing v2:4(Body2) ball bearing v2:4(Body4)]
[S] Bonded279 [ball bearing v2:4(Body2) ball bearing v2:4(Body3)]
[S] Bonded280 [ball bearing v2:4(Body1) ball bearing v2:4(Body10)]
[S] Bonded281 [ball bearing v2:4(Body1) ball bearing v2:4(Body9)]
[S] Bonded282 [ball bearing v2:4(Body1) ball bearing v2:4(Body8)]
[S] Bonded283 [ball bearing v2:4(Body1) ball bearing v2:4(Body7)]
[S] Bonded284 [ball bearing v2:4(Body1) ball bearing v2:4(Body6)]
[S] Bonded285 [ball bearing v2:4(Body1) ball bearing v2:4(Body5)]
[S] Bonded286 [ball bearing v2:4(Body1) ball bearing v2:4(Body4)]
[S] Bonded287 [ball bearing v2:4(Body1) ball bearing v2:4(Body3)]
[S] Bonded288 [Frame 1.2 v1:3 coupler v1:3]
[S] Bonded289 [Frame 1.2 v1:3 top:1]
[S] Bonded290 [slider v4:2 ball bearing v2:4(Body2)]
[S] Bonded291 [slider v4:2 ball bearing v2:4(Body1)]
[S] Bonded292 [slider v4:2 Frame 1.2 v1:3]
[S] Bonded293 [slider v4:2 ball bearing v2:4(Body1)]
[S] Bonded294 [slider v4:2 traveller v8:2]
[S] Bonded295 [rod v1:2 traveller v8:2]

[S] Bonded296 [rod v1:2 Frame 1.2 v1:3]
[S] Bonded297 [rod v1:4 traveller v8:2]
[S] Bonded298 [rod v1:4 Frame 1.2 v1:3]
[S] Bonded299 [Frame 1 v1:2 rod v1:2]
[S] Bonded300 [Frame 1 v1:2 rod v1:2]
[S] Bonded301 [Frame 1 v1:2 rod v1:4]
[S] Bonded302 [Frame 1 v1:2 rod v1:4]
[S] Bonded303 [bolt v1:11 Frame 1 v1:2]
[S] Bonded304 [bolt v1:11 Frame 1 v1:2]
[S] Bonded305 [bolt v1:11 Frame 1 v1:2]
[S] Bonded306 [bolt v1:10 Frame 1 v1:2]
[S] Bonded307 [bolt v1:10 Frame 1 v1:2]
[S] Bonded308 [bolt v1:10 Frame 1 v1:2]
[S] Bonded309 [bolt v1:2 Frame 1 v1:2]
[S] Bonded310 [bolt v1:2 Frame 1 v1:2]
[S] Bonded311 [bolt v1:2 Frame 1 v1:2]
[S] Bonded312 [bolt v1:16 Frame 1 v1:2]
[S] Bonded313 [bolt v1:16 Frame 1 v1:2]
[S] Bonded314 [bolt v1:16 Frame 1 v1:2]
[S] Bonded315 [ball bearing v2:2(Body11) ball bearing v2:2(Body12)]
[S] Bonded316 [ball bearing v2:2(Body11) ball bearing v2:2(Body12)]
[S] Bonded317 [ball bearing v2:2(Body11) ball bearing v2:2(Body12)]
[S] Bonded318 [ball bearing v2:2(Body11) ball bearing v2:2(Body12)]
[S] Bonded319 [ball bearing v2:2(Body11) ball bearing v2:2(Body12)]
[S] Bonded320 [ball bearing v2:2(Body11) ball bearing v2:2(Body12)]
[S] Bonded321 [ball bearing v2:2(Body11) ball bearing v2:2(Body12)]
[S] Bonded322 [ball bearing v2:2(Body11) ball bearing v2:2(Body12)]
[S] Bonded323 [ball bearing v2:2(Body10) ball bearing v2:2(Body12)]
[S] Bonded324 [ball bearing v2:2(Body10) ball bearing v2:2(Body11)]
[S] Bonded325 [ball bearing v2:2(Body9) ball bearing v2:2(Body12)]
[S] Bonded326 [ball bearing v2:2(Body9) ball bearing v2:2(Body11)]
[S] Bonded327 [ball bearing v2:2(Body8) ball bearing v2:2(Body12)]
[S] Bonded328 [ball bearing v2:2(Body8) ball bearing v2:2(Body11)]
[S] Bonded329 [ball bearing v2:2(Body7) ball bearing v2:2(Body12)]
[S] Bonded330 [ball bearing v2:2(Body7) ball bearing v2:2(Body11)]
[S] Bonded331 [ball bearing v2:2(Body6) ball bearing v2:2(Body12)]
[S] Bonded332 [ball bearing v2:2(Body6) ball bearing v2:2(Body11)]
[S] Bonded333 [ball bearing v2:2(Body5) ball bearing v2:2(Body12)]
[S] Bonded334 [ball bearing v2:2(Body5) ball bearing v2:2(Body11)]
[S] Bonded335 [ball bearing v2:2(Body4) ball bearing v2:2(Body12)]
[S] Bonded336 [ball bearing v2:2(Body4) ball bearing v2:2(Body11)]
[S] Bonded337 [ball bearing v2:2(Body3) ball bearing v2:2(Body12)]
[S] Bonded338 [ball bearing v2:2(Body3) ball bearing v2:2(Body11)]
[S] Bonded339 [ball bearing v2:2(Body2) Frame 1 v1:2]
[S] Bonded340 [ball bearing v2:2(Body2) ball bearing v2:2(Body10)]
[S] Bonded341 [ball bearing v2:2(Body2) ball bearing v2:2(Body9)]
[S] Bonded342 [ball bearing v2:2(Body2) ball bearing v2:2(Body8)]
[S] Bonded343 [ball bearing v2:2(Body2) ball bearing v2:2(Body7)]
[S] Bonded344 [ball bearing v2:2(Body2) ball bearing v2:2(Body6)]
[S] Bonded345 [ball bearing v2:2(Body2) ball bearing v2:2(Body5)]
[S] Bonded346 [ball bearing v2:2(Body2) ball bearing v2:2(Body4)]
[S] Bonded347 [ball bearing v2:2(Body2) ball bearing v2:2(Body3)]
[S] Bonded348 [ball bearing v2:2(Body1) ball bearing v2:2(Body10)]

[S] Bonded349 [ball bearing v2:2(Body1) ball bearing v2:2(Body9)]
[S] Bonded350 [ball bearing v2:2(Body1) ball bearing v2:2(Body8)]
[S] Bonded351 [ball bearing v2:2(Body1) ball bearing v2:2(Body7)]
[S] Bonded352 [ball bearing v2:2(Body1) ball bearing v2:2(Body6)]
[S] Bonded353 [ball bearing v2:2(Body1) ball bearing v2:2(Body5)]
[S] Bonded354 [ball bearing v2:2(Body1) ball bearing v2:2(Body4)]
[S] Bonded355 [ball bearing v2:2(Body1) ball bearing v2:2(Body3)]
[S] Bonded356 [ball bearing v2:2(Body1) slider v4:2]
[S] Bonded357 [ball bearing v2:2(Body1) slider v4:2]
[S] Bonded360 [rail1 Copy v4:2 traveller v8:3]
[S] Bonded361 [rail1 Copy v4:2 Frame 1.2 v1:3]
[S] Bonded362 [rail1 Copy v4:2 Frame 1 v1:2]
[S] Bonded363 [rail1 Copy v4:2 traveller v8:3]
[S] Bonded366 [rail1 Copy v4:2 traveller v8:3]
[S] Bonded369 [rail1 Copy v4:2 traveller v8:3]
[S] Bonded370 [rail1 Copy v4:2 traveller v8:3]
[S] Bonded371 [rail1 Copy v4:2 Frame 1.2 v1:3]
[S] Bonded372 [rail1 Copy v4:2 Frame 1.2 v1:3]
[S] Bonded373 [rail1 Copy v4:2 Frame 1.2 v1:3]
[S] Bonded374 [rail1 Copy v4:2 Frame 1.2 v1:3]
[S] Bonded375 [rail1 Copy v4:2 bolt v1:16]
[S] Bonded376 [rail1 Copy v4:2 bolt v1:12]
[S] Bonded377 [rail1 Copy v4:2 bolt v1:11]
[S] Bonded378 [rail1 Copy v4:2 bolt v1:15]
[S] Bonded379 [rail1 Copy v4:2 bolt v1:2]
[S] Bonded380 [rail1 Copy v4:2 bolt v1:13]
[S] Bonded381 [rail1 Copy v4:2 bolt v1:10]
[S] Bonded382 [rail1 Copy v4:2 bolt v1:14]
[S] Bonded391 [Frame 2 v1:2 Frame 1.2 v1:3]
[S] Bonded392 [Frame 2 v1:2 rail1 Copy v4:2]
[S] Bonded393 [Frame 2 v1:2 top:1]
[S] Bonded394 [Frame 2 v1:2 bolt v1:15]
[S] Bonded395 [Frame 2 v1:2 bolt v1:12]
[S] Bonded396 [Frame 2 v1:2 bolt v1:14]
[S] Bonded397 [Frame 2 v1:2 bolt v1:13]
[S] Bonded398 [Frame 2 v1:2 top:1]
[S] Bonded399 [Frame 2 v1:2 rod v1:4]
[S] Bonded400 [Frame 2 v1:2 rod v1:4]
[S] Bonded401 [Frame 2 v1:2 rod v1:2]
[S] Bonded402 [Frame 2 v1:2 rod v1:2]
[S] Bonded403 [Frame 2 v1:2 bolt v1:15]
[S] Bonded404 [Frame 2 v1:2 bolt v1:15]
[S] Bonded405 [Frame 2 v1:2 bolt v1:12]
[S] Bonded406 [Frame 2 v1:2 bolt v1:12]
[S] Bonded407 [Frame 2 v1:2 bolt v1:14]
[S] Bonded408 [Frame 2 v1:2 bolt v1:14]
[S] Bonded409 [Frame 2 v1:2 bolt v1:13]
[S] Bonded410 [Frame 2 v1:2 bolt v1:13]

Mesh

Type	Nodes	Elements
Solids	1670757	1061905

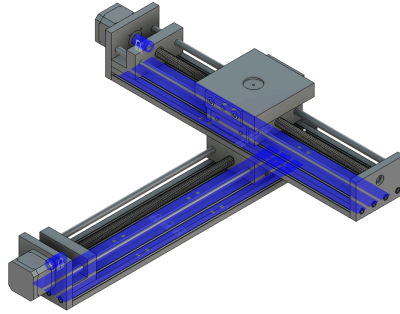
▣ Load Case1

▣ Constraints

▣ Fixed4

Type	Fixed
Ux	Fixed
Uy	Fixed
Uz	Fixed

▣ Selected Entities

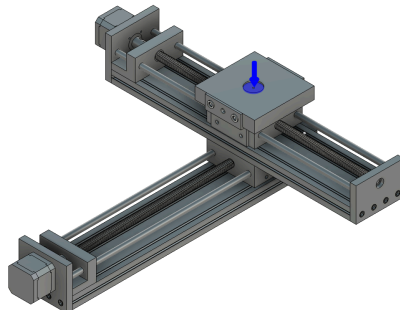


▣ Loads

▣ Force2

Type	Force
Magnitude	20.00 N
X Value	0.00 N
Y Value	-20.00 N
Z Value	0.00 N
Force Per Entity	No

▣ Selected Entities



▣ Results

▣ Result Summary

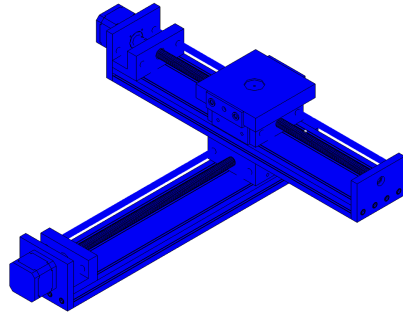
Name	Minimum	Maximum
Safety Factor		

Safety Factor (Per Body)	15.00	15.00
Stress		
von Mises	0.00 MPa	12.787 MPa
1st Principal	-3.528 MPa	15.436 MPa
3rd Principal	-13.642 MPa	3.446 MPa
Normal XX	-13.596 MPa	15.168 MPa
Normal YY	-4.913 MPa	4.943 MPa
Normal ZZ	-4.04 MPa	4.007 MPa
Shear XY	-4.594 MPa	3.178 MPa
Shear YZ	-1.351 MPa	1.693 MPa
Shear ZX	-2.305 MPa	3.025 MPa
Displacement		
Total	0.00 mm	0.016 mm
X	-9.501E-04 mm	0.001 mm
Y	-0.016 mm	8.929E-04 mm
Z	-7.135E-04 mm	3.843E-04 mm
Reaction Force		
Total	0.00 N	27.505 N
X	-27.013 N	14.805 N
Y	-3.972 N	5.031 N
Z	-6.751 N	5.18 N
Strain		
Equivalent	0.00	7.432E-05
1st Principal	-2.050E-07	8.220E-05
3rd Principal	-6.344E-05	1.048E-07
Normal XX	-5.968E-05	6.593E-05
Normal YY	-2.036E-05	1.885E-05
Normal ZZ	-1.152E-05	9.940E-06
Shear XY	-5.645E-05	3.905E-05
Shear YZ	-1.660E-05	2.081E-05
Shear ZX	-2.833E-05	3.718E-05
Contact Pressure		
Total	0.00 MPa	7.978 MPa
X	-5.502 MPa	7.902 MPa
Y	-4.997 MPa	3.377 MPa
Z	-1.90 MPa	1.744 MPa
Contact Force		
Total	0.00 N	27.437 N
X	-26.954 N	19.597 N
Y	-5.608 N	6.606 N
Z	-6.53 N	5.121 N

☐ Safety Factor

☐ Safety Factor (Per Body)

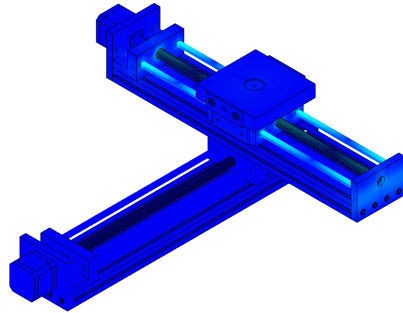
0.00  8.00



Stress

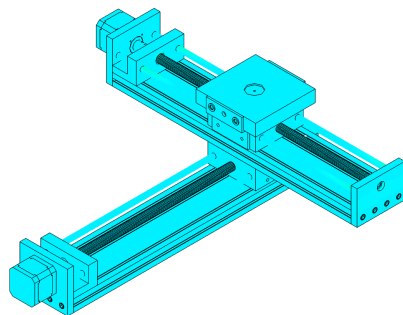
von Mises

[MPa] 0.00 12.787



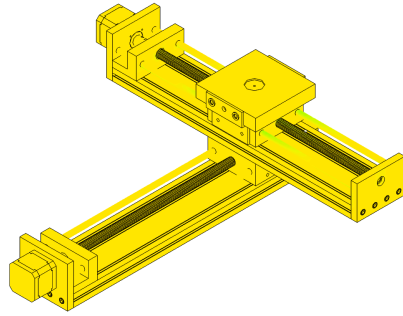
1st Principal

[MPa] -3.528 15.436



3rd Principal

[MPa] -13.642 3.446



▣ Displacement

▣ Total

[mm] 0.00  0.016

